

STIZK3993 ACADEMIC PROJECT 1

Project Design

< SECURE PRIVATE CLOUD STORAGE ON LINUX SERVER/PLATFORM >

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1.0 INTRODUCTION

1.1 PURPOSE

This Secure Cloud Storage (SCS) is required to store and safeguard exam papers for the current semester, as well as to provide a secure means of ensuring that the system is protected from unauthorized users. To make this system possible, it will be used as a centralized system that will serve as the storage for all exam papers for each course at the school of computing. This system will enable the cloud flare zero trust feature, which uses zero trust network access to control authorized and authenticated users over a set period to ensure that the user gaining access is who they claim to be by continuing the authentication process.

1.2 SCOPE

The SCS will be used to store exam papers for the school of computing's current semester. This system will use zero trust network access functionality to control system access. Only users with an authorized connection will be able to access the system. This system will also include functions such as login and registration, allowing users to create and log into their own system accounts. The user's ability to upload and download exam papers is determined by the permissions set on their account. This system will also require 2FA for any actions such as downloading, deleting, or uploading exam papers to the system.

However, it will not be covered in this paper about 2FA. Functionality and the ability to view exam papers in the system will not be implemented due to security concerns that may arise. Administrators will assign users access to specific courses and folders. This system will also include backup functionality, allowing you to store backup folders for each exam paper. Aside from that, encryption will be used to protect the file from unauthorized access.

1.3 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

[List all specific/domain terms used in this document.]

Example:

ZTNA -	Zero trust network access (ZTNA), also known as the software-defined perimeter (SDP), is a set of technologies and functionalities that enable secure access to internal applications for remote users.
WARP -	The Cloudflare WARP client allows individuals to have a faster, more secure, and more private experience online. The WARP client sits between your device and the Internet and has several connection modes to better suit different needs.
Cloudflared	Cloudflare is among the world's largest networks. Businesses, non-profits, bloggers, and anyone with an Internet presence now have faster, more secure websites and apps thanks to Cloudflare.

2.0 GENERAL DESCRIPTION

This section provides a high-level description of *Secure Cloud Storage (SCS)*, identifies the users involved and their roles, and specifies the general constraints, assumptions and dependencies (if any).

2.1 SYSTEM FUNCTIONS

The SCS will be used to store exam papers for the School of Computing. This system will focus on security measures to ensure that only authorized users have access to it due to the confidentiality of the exam papers. This system will include several critical functions to ensure that it functions properly and meets the needs of the users. The system's functions are as follows:

- **Zero trust network access:** Cloudflare Zero Trust offers a zero-trust platform that can be used. Zero trust network access will be used to restrict system access using Cloudflare's virtual private network, or WARP, which allows system access to only users who are connected to the organization.
- **Account management:** To have a second layer of access control over which folders users can access, functions like login and registration will be implemented. Other feature will be available to administrators, assisting them in managing user accounts on courses that have been assigned to them as well as adding and removing users.
- **File management:** With this feature, users can download and upload exam papers to cloud storage, among other things. According to the user account settings, the user will be able to choose which courses' exam papers will be uploaded.
- **File encryption:** to preserve the integrity of the file, this feature will be activated and will automatically encrypt exam papers stored in secure cloud storage.
- **Backup:** an exam paper backup that can be scheduled for specific times.

2.2 USER

This system will be used by three user groups which is consist of System administrator, coordinator and lecturer. The **system administrator** will oversee allocating courses to coordinators and lecturer. Courses can be assigned to lecturer by this system administrator, who can also designate a coordinator for the course. Permission to view all courses and perform all necessary system actions is granted to the system administrator. Other than that, system administrator able to do a backup or recovery if any error happens to the system. System administrator also able to view all account that has been register, make any necessary change to the account if needed and approve any account that register .

The second group is the coordinator is responsible on managing exam paper for certain course, this user able to add and remove regular users to the course and has the same permissions as regular users. The final group is lecturer role who will be responsible on creating a part or section of the exam paper based on coordinator needed, this user requires to assign to courses first before it can upload and download exam papers based on the permissions set.

2.3 GENERAL CONSTRAINTS

Software limitation:

- *Ubuntu LTS as operating system*
- *Limited to functional of cloud flared zero trust normal functionality.*
- *Limited to programming language html , CSS , JavaScript and PHP.*

Hardware limitation:

- *PC capable of running browser to open Secure cloud storage.*
- *PC capable of running cloud flared warp applications.*

2.4 ASSUMPTIONS AND DEPENDENCIES

Assumption:

- Users have adequate skill to use computer and the software.
- Users have basic knowledge on cyber security awareness.

Dependencies:

- System is dependent on the Cloudflare zero trust to operate and go online.

3.0 REQUIREMENT

This section contains the listing of all functional and non-functional requirements for Secure , the use case(s) model, analysis object model as well as dynamic model. An initial interface prototype is also included at the end of this section.

Both functional and non-functional requirements are as shown in Table 1 and 2 respectively. Each requirement is either a mandatory (**M**) or desirable (**D**) or an optional (**O**).

M – Requirement that the system must do

D – Requirement that the system preferably do

O – Requirement that the system may do

Example:

TABLE 1 Functional Requirement List

No.	Requirement	Priority
1.	To implement a Login and Registration System for user authentication and account creation.	
1.1	To allow users to create accounts by providing necessary details, such as email, password, and name.	M
1.2	To ensure secure user authentication by validating login credentials (email and password).	M
1.3	To provide forget password option to users who forget their credentials using email.	D
1.4	To allow administrators to approve or deny user registrations.	M
1.5	To ensure all action such as login and register are logged for auditing and security purpose.	O
2.	To implement Account management to allow admin to control and manage account	
2.1	To allow admin to manage account roles such as assigning role for account	M
2.2	To allow admin to edit account details such as email, name or any details that is necessary.	M
2.3	To allow admin delete account that no longer needed.	M
3	To implement File management that will handling function such as download, upload and file management.	
3.1	To allow user with require roles to upload, download and delete file into exam paper directory that has been assigned to user.	M
3.2	To allow coordinator to view and manage role for course that they assign to and perform such action such as add, remove and view user that been register to certain course.	M
3.3	To ensure all action such as upload, download and delete perform by any user is being logged for auditing and security purpose.	D
4	To implement zero trust network access (ZTNA) to control access to system	
4.1	To allow user to connect to ZTNA VPN using WARP application both via mobile and PC.	M
4.2	To ensure user with ZTNA VPN only able to access the system	M

- 5 To allow admin to backup and recovery exam paper directory**
- 5.1 To allow admin to view and backup existing exam paper in cloud storage *D*
- 5.2 To allow admin able to recovery exam paper data in case of any error or data wipe happen. *D*

TABLE 2 Non-functional Requirement List

No.	Requirement	Priority
1.	To allow file encryption automatically.	
1.1	All file store in secure cloud storage will be encrypted to maintain confidential	<i>M</i>
1.2	The system must ensure that encrypted files cannot be accessed by unauthorized users.	<i>D</i>
2.		
2.1		...

The above requirements are gathered using the following techniques:
 [List the techniques you used to gather the requirement.]

3.1 SYSTEM MODEL

- The main page will be displayed when user open Secure cloud storage, and it will show login page for user to login into system.
- Users sign up for a new account as registered user.
- After login successfully, Main page of system will be displayed, item such as Course assign and user role will be displayed on main page of the system.
- User with admin role will be able to view other menu such as Account management in this menu account management such as account approval, add role, edit account details and delete account can be perform in this menu.
- Admin will be able to view all action that be done in system via Log menu.
- Admin will be able to backup and recover data in system like exam paper in Backup and Recovery Menu.
- User with coordinator role will be able to view menu such as File management with permissions to assign course to user account. This coordinator role also will be inheritance all permissions granted to lecturer role.
- User with lecturer account with course assigned will be able to upload and download exam paper depending on directory that they allow to access.

1.1 USE CASE MODEL

Table 3 shows the list of use case(s) associated to each requirement, and their ID.

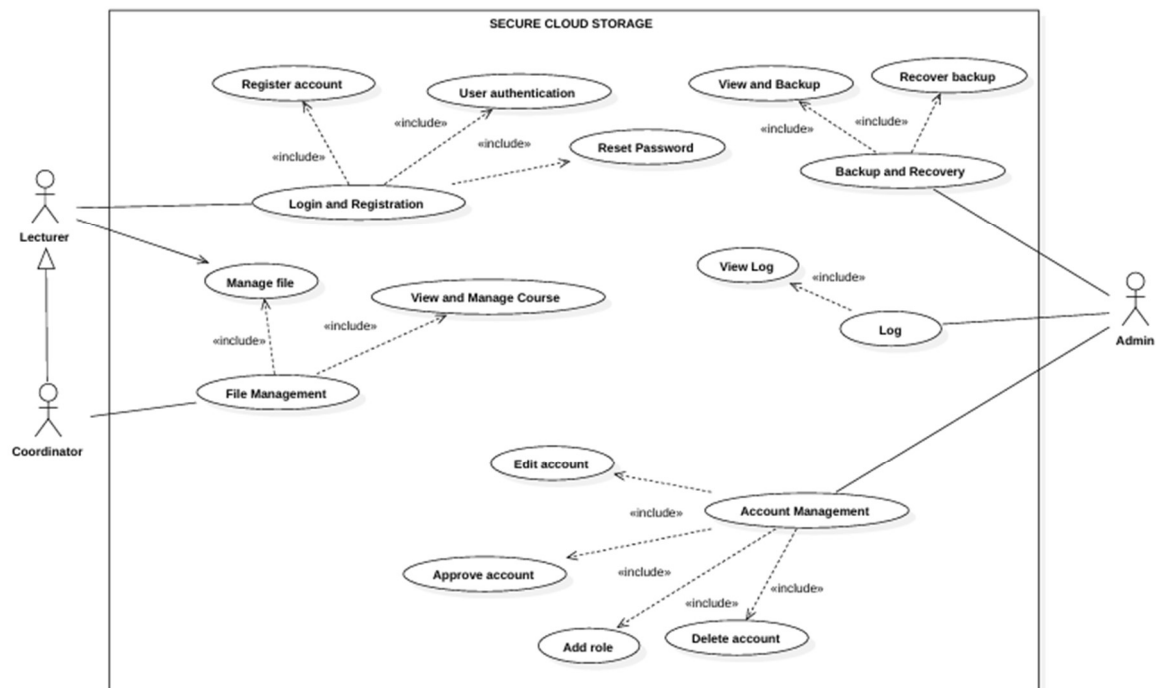
Example:

TABLE 3 Use Case List

No.	Requirement	Use Case(s)	Use Case ID
Login and registration			
1	To allow users to create accounts by providing necessary details, such as email, password, and name.	Register account	AG_01_01

2	To ensure secure user authentication by validating login credentials (email and password).	User authentication	AG_01_02
3	To provide forget password option to users who forget their credentials using email.	Reset Password	AG_01_03
File management			
4	To allow coordinator to view and manage role for course that they assign to and perform such action such as add, remove and view user that been register to certain course.	View and Manage course	AG_01_04
5	To allow user with require roles to upload, download and delete file into exam paper directory that has been assigned to user.	Manage File	AG_01_05
Account management			
6	To allow admin to edit account details such as email, name or any details that is necessary.	Edit account	AG_01_06
7	To allow administrators to approve or deny user registrations.	Approve account	AG_01_07
8	To allow admin to manage account roles such as assigning role for account	Add role	AG_01_08
9	To allow admin delete account that no longer needed.	Delete account	AG_01_09
Log			
10	To ensure all action such as login and register are logged for auditing and security purpose. To ensure all action such as upload, download and delete perform by any user is being logged for auditing and security purpose.	View Log	AG_01_10
Backup and recover			
11	To allow admin to view and backup existing exam paper in cloud storage	View and Backup	AG_01_11
12	To allow admin able to recovery exam paper data in case of any error or data wipe happen.	Recover backup	AG_01_12
Zero trust network access (ZTNA)			
13	To allow user to connect to ZTNA VPN using WARP application both via mobile and PC. To ensure user with ZTNA VPN only able to access the system	Connect to warp	AG_01_13

Secure Cloud storage's usecase



Zero trust network access (WARP)'s Usecase

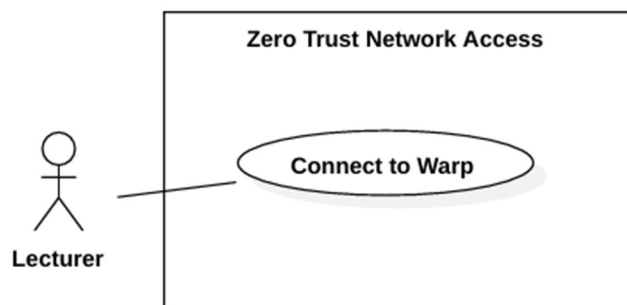


TABLE 4 Use Case Descriptions

Use Case	Descriptions
Register account	This use case allows user without any account to create account in system by entering details such as email, password and username to be register in system. After registration the account will need to be approve by admin. Input: username, email and password. Process: After users enter these details, it will be check if account with same email exists if not the account will be created at same time password will be check if it meets the requirement of password. Exception 1: If the first criteria didn't meet it will display email already exist. Please use another email or forget password. Exception 2: If it didn't meet the second criteria which is password strength it will display "Password must be at least 8 characters long and include an uppercase letter, lowercase letter, and special character." Success: it will display message "Account successfully creates. Please wait for admin to approve your account" and account creation will be log into system log.

<i>User authentication</i>	This use case will be by user with account to login into system. Only user with register email and correct password will have access to system. • INPUT: email, password. • PROCESS: after user click on login button with input details above system will be check with record of account in database if it matches user will be authentication and will be redirect to system page. Else it will be displaying error message and redirect back to login page. • Exception 1: if email or password didn't match with system record it will display "Your email or password is incorrect. Please try again". • Success: if record match with database it will store session and redirect into system page and log the authentication process into system log.
<i>Reset Password</i>	This use case will be used by user with any role to reset their password. By entering user email, temporary password will be sent to email for user to login and reset password when user successfully login. Input: email • Process: email will be check if it exists, if yes random password will be generated and will be sent to register email. Users require to use the temporary password to login and reset their password. • Exception: If email enter not exist in system. This message will be display "No user with <email> exist. Please try again." • Success: if success. This message will be display "Please check you email inbox for temporary password" and action will be log into system logs.
<i>Manage File</i>	This use case will be use by lecturer role to delete, download and upload file into system. The system will be check if user meet the necessary role or permission to perform action select by user. • Input: file to upload, file to be download or delete • Process: system will check if user account meets the necessary permission to conduct action select by user if meet the action will be done. • Exception: If fail. This message will be display "User do not meet the necessary permission to perform this action" • Success: if success. This message will be display "The action has been successfully performs" and action perform by user will be log into system log.
<i>View and Manage course</i>	This use case will be use by user with coordinator role or above to add lecturer to course that they been assigned to create exam paper for, coordinator able to add and remove lecturer from certain course. • Input: email of lecturer • Process: system will be checking if the email exists, if it exist system will add the email into course. • Exception: This message will be display if task failed "This email does not exist. Please try again". • Success: This message will be display if task successfully "Account with <email> has been added to course." and action will be log into system logs.
<i>Edit account</i>	This use case will be use by admin role to edit account details such as email, username and email. • Input: any details that require to be edit. • Process: System will be check if user perform edit meet the permission if meet the detail of the account will be edited. • Exception: This message will be display if task failed "This action can't not be performs. Please try again". • Success: This message will be display if task successfully "Account with <email> has been edited." and action will be log into system logs.
<i>Approve account</i>	This use case will be use by admin to be approved or reject and account that been register into the system, • Input: admin will select action such as reject or approved

	• Process: account status will be change to inactive or active in database. Email to notify user about account is approved will be send. • Exception: This message will be display if task failed “This action can’t not be performs. Please try again”. • Success: This message will be display if task successfully “Account status has been updated” and action will be log into system logs.
<i>Add role</i>	This use case will be used by admin to manage all user role. Admin will be able to add role into user account or add another admin to user. • Input: role to be add into account • Process: system will check if action can be perform based on permission on current user account. If possible, role will be update into database. • Exception: This message will be display if task failed “This action can’t not be performs. Please try again”. • Success: This message will be display if task successfully “Account status has been updated” and action will be log into system logs.
<i>Delete account</i>	This use case will be used by admin to delete account in system. • Input: account that want to be deleted. • Process: system will check if action can be perform based on permission on current user account. If possible, account will be deleted from database. • Exception: This message will be display if task failed “This action can’t not be performs. Please try again”. • Success: This message will be display if task successfully “Account has been deleted.” and action will be log into system logs.
<i>Log</i>	This use case will give admin to view all action that has been done in system. • Process: system will check if user meet the permission to view logs, if meet the requirement all log details will be retrieve and display • Exception: This message will be display if task failed “This action can’t not be performs with current user permission. Please try again” and user will be redirect to system main page. • Success: The log will be display for admin to view.
<i>View and Backup</i>	This use case will be use by admin to view all backup that has been done and admin able to perform backup. • Process: system will check if user meet the permission to view and backup, if meet the requirement all backups will be show and admin will able to back up the current system. • Exception: This message will be display if task failed “Failed to retrieve backup details” and user will be redirect to system main page. • Success: The backup will be display for admin to view.
<i>Recover backup</i>	This use case will be use by admin to recover all backup that has been done, • Process: system will check if user meet the permission to view and backup, if meet the requirement backup will be done and system will be refresh with backup file. • Exception: This message will be display if task failed “Failed to recover backup” and user will be redirect to system main page. • Success: This message will be display if task successfully “Recover successfully.” and action will be log into system logs.
<i>Connect to warp</i>	This use case will be use by user to connect into VPN to have access into system using WARP Zero trust app. • Input: email and tac code send into email. • Process: open zero trust app to login into organization by entering organization name, user require to enter email that has been whitelist to login into warp app with using tac code that has been send to user email. If user able to access into organization warp user can click connect to connect into WARP network. • Exception: This message will be display if tac code been enter is wrong “This account does not have access.”

- **Success:** User will be redirect into warp app after successfully login into organization warp and able to use the WARP VPN by clicking on connected button.
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1.2 ANALYSIS OBJECT MODEL

[Insert sequence diagram for the system]

1.3 DYNAMIC MODEL

[Insert class diagram for the system]

Declaration

I hereby declare that this submission is **my own work** and to the best of my knowledge it contains no materials previously published or written by another.

Student's Signature

Date

Supervisor's Approval

☐	Approve without modification
☐	Approve with modification
☐	Reject

Remark:

Lecturer's Signature & Stamp

Date